UnderwaterDG: Domain Generalized Semantic Segmentation in Underwater Scenes via Decoupling Visibility and Appearance

Yuhang Zhang, Yi Tang, Shiyu Deng, Zhengyu Zhang, Xin Lin, and Guoguang Hua

***Abstract*—Domain generalized semantic segmentation (DGSS) has made significant progress in urban scenes but remains unexplored in marine environments. Underwater images present diverse appearance and visibility variations caused by water quality, lighting, and habitat conditions, which significantly differ from those in urban scenes, making urban DGSS methods less effective in such environments. To fill this gap, we formally investigate DGSS for underwater scenes, termed UnderwaterDG, and propose a novel method focusing on reducing visibility and appearance domain shift. Specifically, we first propose a visibility–appearance decoupling modeling (VADM) framework that integrates the atmospheric scattering model (ASM) with adaptive instance normalization (AdaIN), enabling the disentanglement of underwater images into appearance and visibility components. Second, we propose a decoupled consistency loss (DCL) that enforces prediction consistency under variations in decoupled appearance and visibility, facilitating the learning of appearance- and visibility-invariant representations. Third, we propose a curriculum-guided style-masking perturbation (CSMP) strategy, which enhances generalization by dynamically modulating the difficulty of the disentanglement process. To the best of our knowledge, this is the first DGSS framework tailored for underwater environments, advancing semantic understanding in open and dynamic marine settings. Extensive experiments demonstrate the effectiveness and generalization capability of the proposed approach across diverse underwater datasets.**

***Index Terms*—Appearance-visibility-invariant representations, decoupling visibility and appearance, domain generalization (DG), underwater semantic segmentation, visual foundation model (VFM).**

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I. Introduction

SEMANTIC segmentation, which assigns a category to each pixel in an image, is one of the core technologies for scene understanding. It has been widely applied in various fields such as autonomous driving [1], [2], intelligent healthcare [3], and remote sensing [4], [5]. Significant progress has been made under the assumption of access to large-scale labeled datasets and distributional consistency between training and testing sets [6]. However, such assumptions rarely hold in real-world scenarios.

To address this limitation, domain generalization (DG) has been proposed, aiming to mitigate the domain shift between the source and target domains and enhance model performance in the target domain [7]. It seeks to learn domain-invariant representations solely from the source domain, without any information about the target distribution [8]. Existing methods are predominantly developed for structured urban scenes, such as Cityscapes [9] and BDD [10].

In contrast, domain generalized semantic segmentation (DGSS) for underwater environments remains largely unexplored, despite its growing relevance in real-world applications, including marine exploration, ecological monitoring, and autonomous underwater robotics [11], [12]. Methods developed for structured urban scenes often rely on assumptions that do not hold in underwater conditions, such as relatively stable lighting, texture-rich objects, and well-defined class boundaries. These differences make it nontrivial to directly transfer urban DGSS methods to underwater scenarios. To support this claim, we also provide a comprehensive analysis below. Given the critical role of scene understanding in assessing biodiversity, managing marine resources, and sustaining fisheries, developing Domain Generalized semantic segmentation for Underwater environments (UnderwaterDG) is not only important but urgently needed.

To highlight the distinctive characteristics of underwater environments compared to urban scenes, several representative examples are presented in Fig. 1. Underwater scenes exhibit substantial diversity in both appearance and visibility due to variations in depth, water quality, lighting, and habitat types. As shown in the second column, scenes range from shallow waters and deep seas to underwater caves, each introducing different environmental challenges. More specifically, the third column illustrates appearance variations in color, texture, and morphology of the same object categories across domains. The fourth column highlights visibility inconsistencies, where

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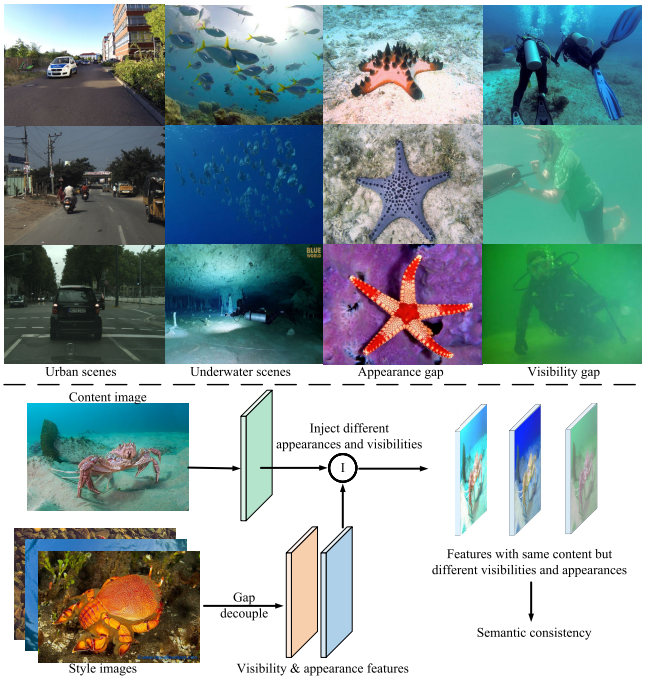


Fig. 1. (Top) Visualization of the domain gap in UnderwaterDG and (bottom) core design of our method.

turbidity, scattering, and low contrast lead to significant differences in image clarity. Therefore, appearance variation and visibility inconsistency jointly contribute to significant domain shifts in underwater environments.

Alternatively, structured urban scenes (first column) predominantly differ in appearance while generally maintaining stable visibility conditions. This fundamental difference implies that DGSS methods tailored to urban settings, often focusing solely on appearance-level adaptation, may fail to generalize in underwater scenarios. Some methods [13], [14] enhance underwater images before segmentation, but since they are not developed under a DG paradigm, the enhancement models often overfit to specific domains and fail in new environments, rendering unreliable segmentation. The above analysis presents the necessity for a tailored solution that explicitly models both appearance and visibility variations.

Thus, we propose a novel UnderwaterDG framework that explicitly decouples appearance and visibility, with a brief overview of its core idea illustrated in Fig. 1. This decoupling addresses the shortcomings of existing DGSS methods in underwater scenarios. While prior methods either focus solely on appearance-level adaptation or treat domain differences as a monolithic factor, they fail to account for the independent variations of visibility (e.g., turbidity and low contrast) and appearance (e.g., color and texture). By disentangling these two factors, our method allows targeted feature augmentation and consistency learning for each component.

In summary, our contributions are concluded as follows.

1) We introduce a new task, UnderwaterDG, focusing on DGSS in underwater scenes. We identify that domain gaps in underwater scenarios are mainly attributed to differences in visibility (e.g., clarity and lighting) and appearance (e.g., color and texture). Furthermore, we analyze the limitations of existing urban-scene DGSS methods when applied to underwater domains.
2) We propose a visibility–appearance decoupling modeling (VADM) method that decouples these two factors. Unlike existing DGSS methods that treat domain difference as a monolithic concept, our disentangled formulation enables fine-grained feature augmentation and consistency learning.
3) We propose a decoupled consistency loss (DCL) and a curriculum-guided style-masking perturbation (CSMP) strategy, which collaboratively learn visibility- and appearance-invariant representations.
4) Extensive experiments demonstrate the effectiveness and generality of our approach across diverse underwater datasets and multiple backbone architectures, validating its robustness and practical value.

II. Related Work

A. Underwater Semantic Segmentation

Underwater semantic segmentation has become a fundamental technique for underwater visual perception and understanding, attracting increasing research interest in recent years due to the growing demand for marine resource exploration [15]. To enhance the contextual feature extraction capability of segmentation models, Liu et al. [16] proposed a multiscale feature fusion method that effectively integrates shallow and deep features, thereby improving the perception of small objects. Qi et al. [17] introduced SGUIE-Net, an underwater segmentation network that incorporates semantic attention with multiscale perception. Their method achieves end-to-end underwater image enhancement while preserving texture details and improving adaptability to various types of degradation. In addition, with the expanding application of large models in computer vision, Xu et al. [18] and Hong et al. [19] fine-tuned large-scale segmentation models to adapt them to underwater environments, offering more powerful foundational support for underwater visual understanding.

Due to common issues in underwater imagery, such as low contrast, color distortion, and blurring [20], many studies have treated underwater image enhancement as a crucial preprocessing step for semantic segmentation. Liu et al. [13] proposed a dual-adversarial contrastive learning-based enhancement method that mitigates uncertainty in the enhancement process through bidirectional mapping, thereby improving both image quality and model generalization. Yu et al. [14] introduced an unsupervised enhancement approach based on decomposable representations, which separates task-relevant features from distortion features and integrates a semantic-aware contrastive module to enhance semantic expressiveness in the restored images.

However, these methods often fail across different underwater scenes, as they typically rely on distributional assumptions specific to a single environment and lack the ability to generalize across variations in visibility and appearance.

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B. Cross-Domain Semantic Segmentation for Urban Scenes

Cross-domain semantic segmentation generally includes two major settings: domain adaptation semantic segmentation (DASS) and DGSS. In DASS, recent works [21], [22], [23] have improved adaptation stability and pseudolabel reliability under domain shift. For instance, Zhao et al. [24] improve online self-training stability by introducing a dynamic teacher update mechanism and a training-consistency-based resampling strategy. SND [25] improves pseudolabel reliability through stability-guided denoising, where stable neighbors are retrieved to refine unstable samples, effectively reducing class, domain, and confirmation biases during source-free adaptation. SeCo [26] enhances pseudolabel quality by formulating supervision at the semantic-connectivity level, enabling robust aggregation and correction of speckled closed- and open-set noise, further extended in SeCoV2 [27] with uncertainty-aware correction and broader generalization. However, DASS is limited to adaptation within a single known target domain, motivating the need for DGSS.

DGSS methods for urban scenes have seen significant development and can generally be categorized into style extension, style removal, and domain-invariant representation learning [28]. These strategies are not mutually exclusive and are often integrated to achieve superior generalization performance [29]. Style extension methods try to generate images with diverse styles for covering possible unseen domains, such as WildNet [30], SHADE [31], and FSDR [32]. DGInStyle [33] further leverages semantically controlled diffusion models to generate style-diverse training data, including night, rain, sunset, and other diverse scenarios and styles. Style removal methods aim to eliminate domain-specific information from deep representation by using feature normalization, whitening, and so on, such as IBN [34], ISW [35], and BlindNet [36]. Representation learning approaches, such as PPAR [37], DPCL [38], and FGSS [39], focus on learning domain-invariant and domain-stable features through various strategies, such as contrastive learning and classwise prototype alignment.

Recently, with the remarkable generalization capabilities of visual foundation models (VFMs), several studies have explored adapting them to semantic segmentation by introducing additional task-specific parameters and performing fine-tuning, such as Rein [40], FADA [41], and CDG [42]. FisherTune [43] proposes a Fisher-information-guided robust tuning strategy to preserve the generalization capability of VFMs.

Meanwhile, several methods have focused on improving generalization for multitask adaptability and adverse weather conditions [41], [44], [45]. For example, HarDiff [46] leverages low-frequency structural priors and high-frequency target details to enhance the generalization of diffusion models across multiple dense vision tasks, such as semantic segmentation, depth estimation, and haze removal. AdaDCP [45], on the other hand, learns weather-invariant and weather-variant frequency components to effectively mitigate the domain gap from clear scenes to adverse conditions like fog, snow, and nighttime, achieving robust segmentation on unseen weather scenarios.

As discussed above, these methods focus on the urban scenes and generally treat the domain difference as a coarse concept, which may limit their effectiveness in handling the complex variations present in underwater environments.

C. Curriculum Learning

Curriculum learning is a training strategy that presents samples to a model in a meaningful order, typically from easy to hard, to improve learning efficiency and performance [47]. CCDA [48] is a curriculum-style cross-domain segmentation framework that aligns local-to-global features by adapting easy-to-hard target patches. CPP [49] guides the model from easier, object-centric semantic alignments to more complex instances, improving mask precision while mitigating noise and overfocus issues for weakly supervised referring image segmentation. Curri-AFDA [50] is a medical image segmentation framework that progressively transfers semantic knowledge from source to target domains by gradually introducing increasingly challenging amplitude information in the Fourier space. ATP [51] is a source-free domain adaptive semantic segmentation framework that progressively transfers knowledge from easy-to-hard target samples via curriculum-style feature alignment and complementary pseudolabeling to enhance target representations and reduce intradomain discrepancy. These previous effects provide an inspiration for the proposed method.

III. Method

Given a source domain $\mathcal{S}$, the proposed UnderwaterDG task aims to learn a generalized segmentation model by using $\mathcal{S}$'s data and generalize well to multiple unseen target domains $\{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_N\} \in \mathcal{T}$.

The overview of our method is shown in Fig. 2. Our method contains four components: VADM, DCL, CSMP, and supervised learning. The first three modules are newly proposed in this work to explicitly disentangle and regularize visibility and appearance factors across scenes. The last module follows the standard supervised learning pipeline on the source domain and serves as the primary segmentation objective.

For the inference stage, our method requires only the input test image. The segmentation prediction is obtained through a single forward pass of the segmentation model.

A. Preliminaries

The atmospheric scattering model (ASM) is a traditional physical model that is widely used in foggy weather and other scenes that suffer from light scattering and selective absorption, undoubtedly including underwater scenes, which can be defined as

$$I = Jt + A(1 - t) \tag{1}$$

where I, J, A, and t represent the degraded image, ideal high-quality image, atmospheric light constant, and medium transmission diagram, respectively. According to USUIR [52], t quantifies how much scene radiance successfully reaches the camera after attenuation by the medium. As it controls

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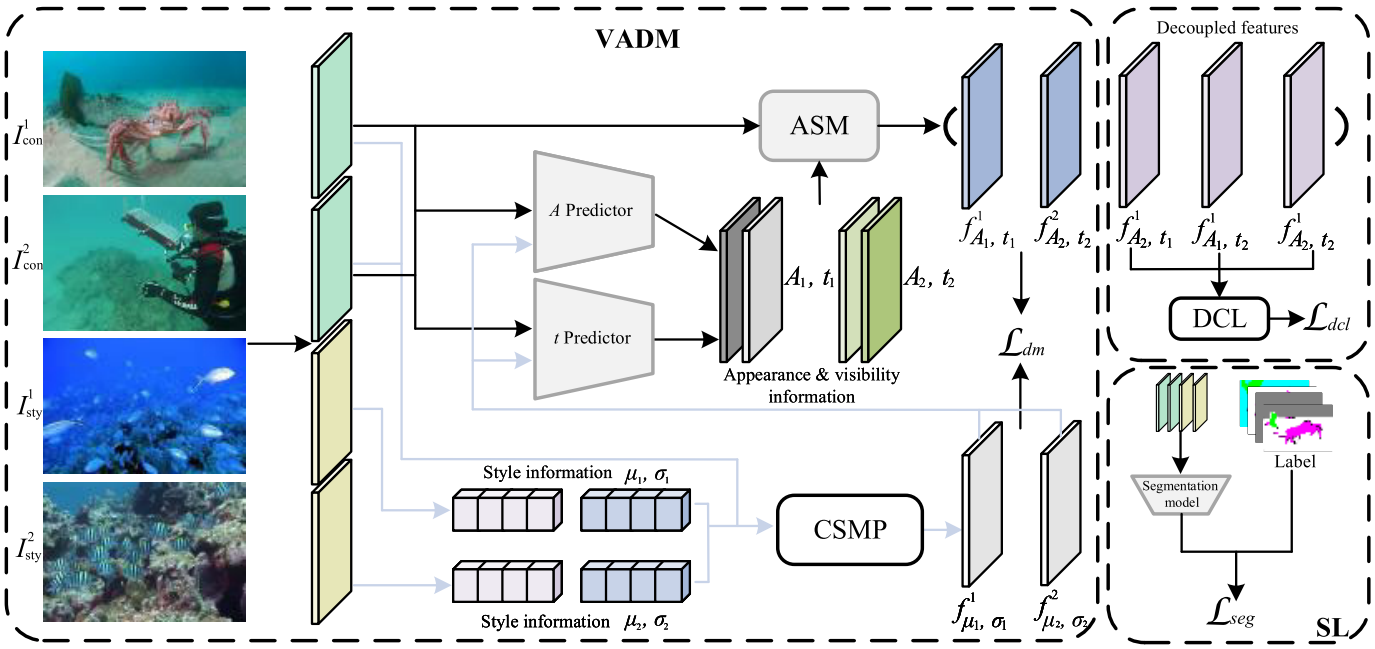


Fig. 2. Overview of our framework. VADM, CSMP, DCL, and SL represent VADM, CSMP, DCL, and supervised learning, respectively.

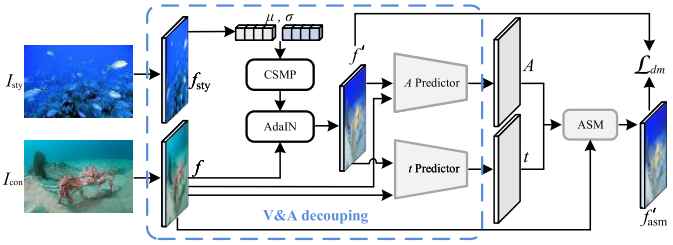


Fig. 3. Illustration of VADM. CSMP and AdaIN generate a stylized feature f' with controlled visibility/appearance changes, while the predictors P_A and P_t extract the corresponding components and reconstruct f'_{asm} through ASM to enforce effective feature decoupling.

the amount of light loss due to scattering and absorption, t naturally corresponds to a visibility factor, reflecting the image's visibility or degree of degradation caused by the environment. Conversely, A represents the ambient background illumination scattered in the medium, which introduces a color cast characteristic of the surrounding water and affects texture extraction, so it can be interpreted as an appearance factor.

B. Visibility–Appearance Decoupling Modeling

To address appearance and visibility shifts in underwater scenes, it is crucial to learn visibility- and appearance-invariant representations. A natural way is to decouple these two factors. Based on this, a widely adopted paradigm, feature augmentation and consistency constraints, can be applied to learn domain-invariant representations, as validated by many DG methods [32], [53], [54]. In our case, the decoupled visibility and appearance features can be recombined with arbitrary samples to augment training data and enforce representation consistency.

To achieve the above target, a VADM method is proposed, as shown in Fig. 3. Although the above description successfully interprets A and t in ASM as appearance and visibility factors, respectively, (1) is primarily designed for underwater image restoration. To adapt ASM to the UnderwaterDG task, (1) can be generalized as the transformation between two images (I and I') that share the same content but differ in visibility and appearance, which can be formulated as

$$I' = It + A(1-t). \tag{2}$$

Since segmentation relies on high-level features rather than images, the above equation can be further reformulated as

$$f' = ft + A(1-t) \tag{3}$$

where f and f' are two features that share the same content but differ in visibility and appearance. A and t no longer represent physical atmospheric light and transmission maps but their feature-level counterparts that parameterize appearance and visibility variations.

Then, the key challenge becomes how to construct f' based on f in a way that explicitly injects controllable changes in visibility and appearance. Adaptive instance normalization (AdaIN) [55] is a style transfer technology that is widely used in image translation and domain adaptation/generalization, serving as a suitable method to construct f'. Given one content feature f extracted from content image I_{con} (provide content information) and one style feature f_{sty} extracted from style image I_{sty} (provide style information), the transferred feature f' is defined as

$$f' = \sigma(f_{\text{sty}}) \cdot \frac{f - \mu(f)}{\sigma(f)} + \mu(f_{\text{sty}}) \tag{4}$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ denote the channelwise mean and standard deviation, respectively. Note that f' retains the original semantic content with f, with the visibility and appearance being modified.

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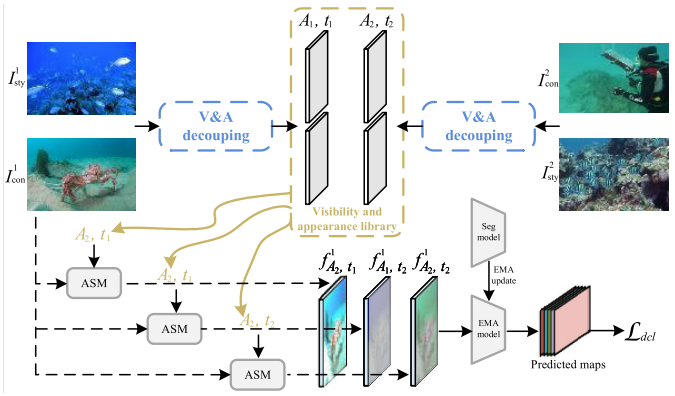


Fig. 4. Illustration of DCL, which applies visibility-only, appearance-only, and combined augmentations with EMA-guided consistency constraints to enforce visibility- and appearance-invariant predictions. V&A decoupling operation is part of VADM, which is shown in Fig. 3. The yellow arrows indicate the different feature variables selected for each augmentation branch. Consistency constraints are enforced via an EMA model derived from the segmentation network.

After that, a A predictor $P_A(\cdot)$ and a t predictor $P_t(\cdot)$ are defined to estimate the transmission (i.e., A and t) between the original feature f and transferred feature f'. To train the predictors $P_A(\cdot)$ and $P_t(\cdot)$ to accurately extract appearance and visibility components from the feature pair (f, f'), a decoupling modeling loss is introduced

$$\mathcal{L}_{dm} = \left|\left|f' - f'_{\text{asm}}\right)\right|_1 \tag{5}$$

where $\hat{t} = P_t(f, f')$ and $\hat{A} = P_A(f, f')$. f'_{asm} can be obtained by ASM, i.e., $f\hat{t} + \hat{A}(1 - \hat{t})$. This loss enforces the predicted appearance and visibility components to faithfully reconstruct the stylized feature f', thereby achieving effective disentanglement.

VADM enables explicit control over feature-level variations and lays a foundation for robust feature augmentation and consistency training in subsequent stages.

C. Decoupled Consistency Loss

To learn visibility- and appearance-invariant representations by leveraging disentangled visibility and appearance information, a DCL is proposed as shown in Fig. 4, consisting of feature augmentation and consistency constraint. The feature augmentation step generates variations in visibility and appearance to simulate domain shifts, while the consistency constraint encourages the model to produce stable and semantically consistent predictions under these variations.

Step 1 (Feature Augmentation): Three types of augmentations are constructed, containing visibility-only, appearance-only, and combined visibility–appearance augmentations. Specifically, visibility variation is controlled by t, while appearance variation is controlled by A.

1) *Visibility augmentation* modifies only the visibility of the original feature.
2) *Appearance augmentation* alters only the appearance (e.g., color/texture).
3) *Visibility–appearance augmentation* modifies both simultaneously.

Specifically, given four features extracted from four different images $(f^1_{\text{ori}}, f^2_{\text{ori}}, f^1_s, f^2_s)$, the transmission pairs (A_1, t_1) and (A_2, t_2) represent the appearance and visibility transformations from the original features f^1_{ori} and f^2_{ori} to the stylized features f^1_s and f^2_s, respectively.

The augmented features are constructed as follows.

1) Visibility augmentation

$$f^1_{A_1,t_2} = f^1_{\text{ori}} t_2 + A_1(1 - t_2). \tag{6}$$

2) Appearance augmentation

$$f^1_{A_2,t_1} = f^1_{\text{ori}} t_1 + A_2(1 - t_1). \tag{7}$$

3) Visibility–appearance augmentation

$$f^1_{A_2,t_2} = f^1_{\text{ori}} t_2 + A_2(1 - t_2). \tag{8}$$

Step 2 (Consistency Constraint): The augmented features $f^1_{A_1,t_2}$, $f^1_{A_2,t_1}$, and $f^1_{A_2,t_2}$ share the same semantic content as the original feature f^1_{ori} but under different visibility and/or appearance conditions. To ensure semantic consistency across these variations, a consistency loss based on the Kullback–Leibler divergence is applied as follows:

$$\begin{aligned}\mathcal{L}_{dcl} = &\text{KL}\left(p(f^1_{\text{ori}}) \,\|\, p_{\text{ema}}(f^1_{A_1,t_2})\right) \\ &+ \text{KL}\left(p(f^1_{\text{ori}}) \,\|\, p_{\text{ema}}(f^1_{A_2,t_1})\right) \\ &+ \text{KL}\left(p(f^1_{\text{ori}}) \,\|\, p_{\text{ema}}(f^1_{A_2,t_2})\right)\end{aligned} \tag{9}$$

where $p(\cdot)$ and $p_{\text{ema}}(\cdot)$ denote the predicted probability distribution of the original segmentation model and the exponential moving average (EMA) model, respectively. The EMA model shares the same structure as the segmentation model and updates each iteration. This loss encourages the model to produce consistent semantic predictions regardless of variations in visibility or appearance.

D. Curriculum-Guided Style-Masking Perturbation

Although VADM enables effective separation of visibility and appearance in a feature, facilitating visibility/appearance augmentation, it inherently relies on the availability of complete and reliable style information.

On the one hand, as the style information used in (4) is generated from a random source sample, despite the diversity brought by conventional data augmentations such as color jitter, the generated styles are still hard to cover that in the target domains, which limits the generalization capability of the proposed DCL.

On the other hand, training the model solely under fully observable style conditions leads to fragile (P_A, P_t) predictors, which struggle to handle partial or degraded style inputs, thereby hindering the learning of robust domain-invariant representations.

To alleviate the issues above, a CSMP strategy is proposed as shown in Fig. 5. Instead of relying solely on full-style statistics (i.e., complete channelwise mean and variance) using (4), CSMP introduces partial masking to the style codes. Given the channelwise mean and standard deviation (μ_s, σ_s) from style feature f_s, a binary mask $M \in \{0, 1\}^C$ is randomly sampled for each channel

$$M_i \sim \text{Bernoulli}(1 - r_{\text{dynamic}}), \quad i = 1, \ldots, C \tag{10}$$

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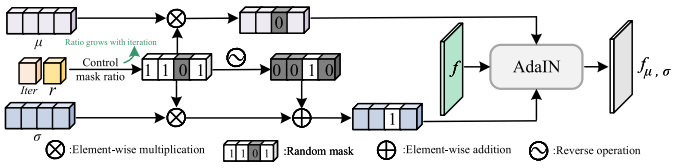


Fig. 5. Overview of CSMP, where curriculum-guided masking is applied to style statistics to simulate incomplete or degraded style conditions, enabling more robust and domain-invariant feature augmentation.

where r_{dynamic} is the dynamic masking ratio that controls the probability of masking a channel; it increases progressively during training in a curriculum manner

$$r_{\text{dynamic}} = \frac{i}{i_{\max}} \cdot r \tag{11}$$

where i and $i_{\max}$ present the current training iteration and max iteration. Then, the masked style statistics $(\tilde{\mu}_s, \tilde{\sigma}_s)$ are defined as

$$\tilde{\mu}_s = M \odot \mu_s, \quad \tilde{\sigma}_s = M \odot \sigma_s + (1 - M). \tag{12}$$

It can be seen that masked channels (where $M_i = 0$) are replaced by zero mean and unit variance. After that, the perturbed style statistics are then used to generate the augmented feature via (4), which is further employed in the VADM through (5).

By randomly masking a proportion of channels in the style feature's mean and variance, the transformed features retain the same content but experience degraded or ambiguous style cues. This simulates real-world conditions where style information may be partially missing due to underwater scattering, absorption, or sensor noise. Our proposed CSMP strategy progressively exposes the model to harder cases, encouraging the model to extract robust and domain-invariant representations under incomplete style conditions.

E. Ultimate Objective

The final training loss of our method is formulated as

$$\mathcal{L} = \mathcal{L}_{seg} + \lambda_1 \mathcal{L}_{dm} + \lambda_2 \mathcal{L}_{dcl} \tag{13}$$

where $\mathcal{L}_{seg}$ is the standard supervised learning objective using cross-entropy loss. $\mathcal{L}_{dm}$ and $\mathcal{L}_{dcl}$ are decoupling modeling loss and DCL, as presented in (5) and (9), respectively.

IV. Theoretical Insights

Our method can be interpreted under the domain adaptation/generalization theories proposed in [56] and further extended to generalizing to unseen domains via the distribution matching (G2DM) framework [57]. Let the original source domain be denoted as $D_s \sim P_s$ and the unseen target domain as $D_{ut} \sim P_{ut}$. During training, the target domain remains inaccessible. To enhance generalization, we apply interpretable augmentation based on visibility and appearance factors, generating a set of diversified source subdomains $D_s^i \sim P_s^i$, where $i \in \{1, \ldots, N_s\}$.

According to G2DM, the target distribution can be approximated by a convex combination of subsource distributions

$$P_{ut} \approx \sum_{i=1}^{N_s} \alpha_i P_s^i, \quad \sum_{i=1}^{N_s} \alpha_i = 1 \tag{14}$$

where α_i are the mixing weights. Define $\hat{P}_{ut}$ as the closest point in the convex hull of the subsource domains to the target distribution. The expected risk on the target domain can be upper bounded as

$$r_{ut}(h) \le \sum_{i=1}^{N_s} \alpha_i r_s^i(h) + \delta + \gamma \tag{15}$$

where $r_s^i(h)$ is the expected risk on each subsource domain. δ represents the discrepancy between subdomains $\{P_s^i\}$. γ measures the gap between the target distribution P_{ut} and its projection $\hat{P}_{ut}$ onto the convex hull of subsource domains, reflecting how well the augmented subdomains cover the target domain's diversity.

Although the bound terms are not directly computable in practice, each component of our training objective is designed as an operational proxy to reduce the corresponding term in the theoretical upper bound. Our method is carefully designed to reduce each of these three components in the following.

1) *Supervised Segmentation Over Subdomains*: The empirical risks $r_s^i(h)$ are approximated by supervised segmentation losses applied to both the original source samples and the constructed visibility-, appearance-, and visibility–appearance-augmented subdomains. By optimizing segmentation performance across these diversified subdomains, the model effectively minimizes the averaged source-domain risk term $\sum_{i=1}^{N_s} \alpha_i r_s^i(h)$.
2) *Cross-Domain Consistency Regularization*: The proposed DCL enforces prediction invariance across subdomains generated from the same semantic content but under different visibility and appearance conditions. This consistency constraint serves as a surrogate mechanism to mitigate the discrepancy between subdomains and is, therefore, conceptually related to the interdomain discrepancy term δ.
3) *Diversity-Driven Subdomain Construction*: The visibility–appearance factor perturbation and the CSMP strategy enlarge the diversity of the constructed subdomains along interpretable directions. This can be viewed as expanding the convex hull spanned by $\{P_s^i\}$, thereby being conceptually associated with the coverage gap term γ, which reflects how well the generated subdomains span potential target-domain variations.

V. Experiment

A. Experimental Setup

1) Datasets: Four underwater datasets are employed to evaluate our method, including SUIM [58], FSUW [59], UVTD [12], and Caveseg [60].

SUIM consists of 1525 training and 110 test RGB images with diverse resolutions (ranging from 1906×1080 to 256×256), collected from oceanic explorations and human–robot

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cooperative experiments across various water types. The dataset integrates scenes from multiple underwater environments, includes images sourced from EUVP [61], USR-248 [62], and UFO-120 [63], and covers eight representative semantic categories. Its wide variation in appearance, illumination, and scene composition makes it a comprehensive benchmark for evaluating underwater semantic segmentation methods.

FSUW, originally designed for few-shot segmentation, contains 29 categories with 461 training and 113 test images. Compared with benchmarks such as SUIM that include only a single coarse animal category, FSUW offers fine-grained underwater species annotations collected from diverse real-world and conservatory scenes, providing a richer testbed for both semantic and few-shot segmentation.

UVTD contains 5380 underwater images with multitask annotations for both low-level (UIE and UIQA) and high-level (USS, UOD, and USOD) tasks, sharing the eight semantic classes of SUIM. Over 10k candidate images were collected from online sources, public datasets, and 100 self-captured videos, and 15 trained volunteers filtered them down to 5380 images that support diverse underwater vision research.

CaveSeg contains 3350 training images and 350 test images collected from underwater cave systems across the USA, Mexico, and Spain, providing pixel-level annotations for 13 semantic categories, including caveline, layered obstacles, open areas, scuba divers, navigation aids, and cave formations. These categories capture critical cues for AUV navigation, such as safe passages, directional markers, diver-aware behaviors, and geometric constraints, making CaveSeg a comprehensive benchmark for semantic segmentation and scene parsing in complex, low-visibility cave environments. Its diverse scenes, high-level annotation detail, and inclusion of rare yet safety-critical classes (e.g., arrows, cookies, and divers) greatly facilitate the development and evaluation of robust visual models for autonomous underwater cave exploration.

To ensure consistency across datasets with different label definitions, all category labels were mapped to align with the SUIM dataset. Specifically, UVTD shares the same eight semantic categories as SUIM and, therefore, requires no relabeling. For Caveseg and FSUW, their original fine-grained or dataset-specific categories were semantically aligned to the closest corresponding SUIM classes (e.g., structural cave elements mapped to sea floor and rocks, and fine-grained marine species grouped into fish and vertebrates or coral and invertebrates).

2) Implementation Detail: Intersection over Union (IoU) is adopted as the primary evaluation metric, and Average mIoU (AmIoU) across all domains is used as the final metric to assess segmentation performance. ResNet-50, MobileNet-V2, DINOv2, and EVA-02 are adopted as the backbones. Specifically, ISW [35] is employed as the baseline when using ResNet-50 and MobileNet-V2, whereas Rein [40] is used when DINOv2 and EVA-02 are adopted. Note that different baselines are selected for different backbones due to the paradigm shift in model scales and training strategies. DGSS methods based on ResNet-50 and MobileNet-V2 typically train all network parameters, while methods using VFMs such as DINOv2 and EVA-02 usually adopt fine-tuning training for partial parameters. Reproducing earlier baselines such as ISW under this new paradigm is nontrivial and may not yield fair or effective comparisons. Therefore, representative baselines are chosen that align with the standard training strategy for each backbone. The training settings are summarized in the following.

ResNet-50 and MobileNet-V2: These models are initialized with ImageNet-pretrained weights. The input resolution is cropped to 768×768, and training is conducted for 40 000 iterations. An SGD optimizer is applied with a learning rate of 0.01, a momentum of 0.9, and a weight decay of 5×10^{-4}. The batch size is set to 4. The visibility–appearance disentanglement is performed at the first stage of the encoder (i.e., layer 1 of the backbone), as prior studies [64] suggest that shallow features mainly encode low-level nonsemantic information, making this stage well-suited for separating degradation-related factors before high-level semantic representations are formed. The same data augmentation is adopted as ISW [35], including color jittering, Gaussian blur, random cropping, horizontal flipping, and random scaling within [0.5, 2.0].

DINOv2 and EVA-02: DINOv2 [65] is initialized with LVD-142M. For EVA-02 [66], the pretrained weights are obtained from a merged dataset consisting of IN-21K, CC12M, CC3M, COCO, ADE20K, Object365, and OpenImages. The input resolution is set to 512×512. Training is conducted for 40 000 iterations using the AdamW optimizer with a learning rate of 0.0001 and a weight decay of 0.05. The batch size is also set to 4. We follow the official Rein [40] training configuration, including random cropping.

Hyperparameter setting: λ_1 and λ_2 are empirically set to 0.0001 and 0.001, respectively. r is set to 0.1 for DINOv2 and EVA02, and 0.4 for Resnet-50 and MobileNet-V2. The momentum coefficient of the EMA model is set to 0.999.

3) Implementation Detail of Predictors P_A and P_t: Given the input pair of features, original feature f, and stylized feature f' (they share the same content but differ in style), their pixelwise difference is calculated

$$\Delta = f - f'. \tag{16}$$

To highlight the most informative regions of variation, an attention mechanism to Δ is applied

$$\text{Attn} = \sigma\left(\text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{Norm}\left(\text{Conv}_{3\times3}(\Delta)\right)\right)\right)\right) \tag{17}$$

where σ denotes the sigmoid activation. This yields an attention-weighted difference map

$$\Delta_{\text{att}} = \Delta \odot \text{Attn}. \tag{18}$$

Next, the original feature, the stylized feature, and the attention-weighted difference are concatenated

$$f_{\text{fused}} = [f \,\|\, f' \,\|\, \Delta_{\text{att}}]. \tag{19}$$

This fused feature encodes both the shared content and the domain shift between f and f'.

Two parallel convolutional branches are employed to estimate A and t from f_{fused}. Each branch consists of a 3×3

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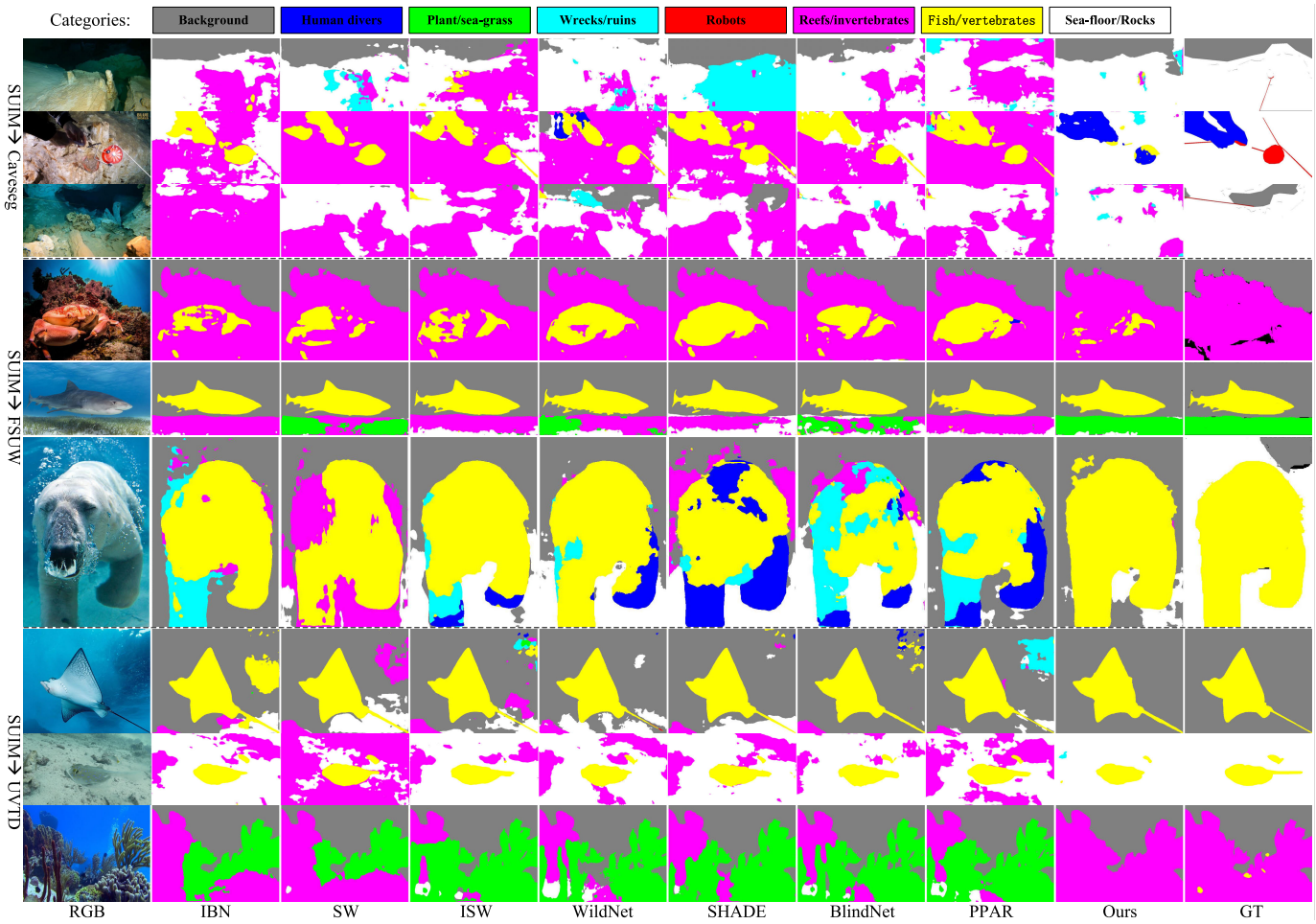


Fig. 6. Visualization comparison with state-of-the-art methods.

convolution followed by normalization and ReLU, and a 1×1 convolution to produce the final output

$$A = \text{Conv}_{1\times1}^{(A)}\left(\text{ReLU}\left(\text{Norm}\left(\text{Conv}_{3\times3}^{(A)}(f_{\text{fused}})\right)\right)\right) \tag{20}$$

$$t = \text{Conv}_{1\times1}^{(t)}\left(\text{ReLU}\left(\text{Norm}\left(\text{Conv}_{3\times3}^{(t)}(f_{\text{fused}})\right)\right)\right). \tag{21}$$

In practice, $A \in \mathbb{R}^{C\times H\times W}$ captures the domain-level appearance transition (e.g., color shift), while $t \in \mathbb{R}^{C\times H\times W}$ models the localized visibility difference.

Note that predictors P_A and P_t are only employed in the training stage. Thus, it will not introduce any resource cost when testing.

B. Comparison With State-of-the-Art Methods

1) Comparison Using ResNet-50 Backbone: ResNet-50 is adopted as the backbone, and seven representative DGSS methods are reproduced for comparison, including ISW, IBN, SW, WildNet, SHADE, BlindNet, and PPAR. Among them, ISW, IBN, SW, and BlindNet are categorized as feature normalization-based methods, while SHADE, WildNet, and PPAR go beyond pure representation learning by additionally incorporating sample expansion strategies to enhance generalization. Cross-domain evaluations are conducted with the following source-to-target configurations: SUIM→{FSUW, Caveseg, UVTD}, Caveseg→{SUIM, FSUW, UVTD}, FSUW→{Caveseg, UVTD, SUIM}, and UVTD→{FSUW, Caveseg, SUIM}. As shown in Table I, our method achieves the highest AmIoU of 35.0%, outperforming the second-best method, WildNet, by 1.2%. Moreover, it achieves either the best or second-best performance in 9 out of 12 cross-domain transfer settings, where each setting corresponds to a distinct source-to-target dataset pair among SUIM, FSUW, Caveseg, and UVTD. Some qualitative results are shown in Fig. 6. As observed, our method shows more complete object segmentations, highlighting its effectiveness.

2) Comparison Using DINOv2 Backbone: To further investigate the performance under VFMs, we also conduct the same experiments using DINOv2. Five representative methods are reproduced, including Rein [40], FADA [41], SET [44], CDG [42], and Causal-Tune [67]. These methods are all based on parameter-efficient fine-tuning (PEFT), where most parameters are frozen, and only a small subset is trainable, which is a widely adopted paradigm in this research field. As shown in Table II, our method achieves an AmIoU of 49.5%, surpassing the second-best method by 1.4%. Moreover, it attains the best or second-best performance in ten out of the 12 generalization settings.

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TABLE I

COMPARISON WITH SOTA METHODS USING RESNET-50 BACKBONE. BEST RESULTS ARE IN BOLD; SECOND-BEST ARE UNDERLINED

Method	Venue	SUIM→			Caveseg→			FSUW→			UVTD→			AmIoU
		FSUW	Caveseg	UVTD	FSUW	SUIM	UVTD	SUIM	Caveseg	UVTD	FSUW	SUIM	Caveseg	
IBN	ECCV'18	32.0	14.3	38.1	9.9	17.5	20.4	41.4	15.1	29.7	40.0	75.5	32.9	30.6
SW	ICCV'19	**40.3**	16.5	39.4	**21.8**	16.0	20.1	40.6	14.5	31.9	**42.3**	74.8	30.3	32.4
ISW	CVPR'21	28.5	22.8	39.5	10.6	18.2	20.7	41.0	13.8	30.7	40.6	73.3	34.4	31.2
WildNet	CVPR'22	32.1	29.4	41.8	11.0	18.8	22.0	**42.3**	20.7	32.3	40.2	**77.5**	37.4	33.8
SHADE	IJCV'24	32.1	28.6	42.0	9.5	18.8	20.8	41.1	**23.4**	**33.1**	38.4	75.1	36.6	33.4
BlindNet	CVPR'24	31.5	17.1	**48.5**	8.7	16.1	19.0	41.3	17.1	31.5	39.4	72.9	35.3	31.5
PPAR	TITS'25	33.4	26.2	46.1	13.0	18.6	20.2	40.2	22.0	31.3	38.8	73.5	34.8	33.4
Ours	-	39.6	**32.7**	41.5	13.7	**22.8**	**23.7**	41.6	20.5	32.3	40.6	72.2	**38.9**	**35.0**

TABLE II

COMPARISON WITH SOTA METHODS USING DINOV2 BACKBONE. BEST RESULTS ARE IN BOLD; SECOND-BEST ARE UNDERLINED

Method	Venue	SUIM→			Caveseg→			FSUW→			UVTD→			AmIoU
		FSUW	Caveseg	UVTD	UVTD	FSUW	SUIM	UVTD	Caveseg	SUIM	FSUW	Caveseg	SUIM	
Rein	CVPR'24	**55.4**	44.2	**58.9**	30.4	11.5	27.6	47.7	49.6	47.4	58.9	63.0	74.5	47.4
FADA	NIPS'24	49.2	36.0	57.3	31.2	11.1	28.4	49.0	48.9	48.0	53.0	**66.1**	62.7	45.1
SET	ACM MM'24	51.3	49.0	58.4	34.3	12.3	31.4	48.0	46.7	**57.8**	56.8	57.7	73.1	48.1
CDG	PR'25	41.8	40.2	58.2	33.2	13.2	28.7	47.3	48.1	48.4	46.4	60.0	67.8	44.4
Causal-Tune	AAAI'26	52.6	43.9	51.9	34.3	12.8	30.1	46.3	46.6	45.2	57.3	51.5	71.9	45.4
Ours	-	50.4	**52.3**	**58.9**	**35.8**	**14.6**	**33.1**	**49.3**	**51.0**	48.9	**64.3**	60.2	**75.3**	**49.5**

TABLE III

COMPARISON WITH SOTA METHODS USING EVA02 BACKBONE. BEST RESULTS ARE IN BOLD; SECOND-BEST ARE UNDERLINED

Method	Venue	SUIM→			Caveseg→			FSUW→			UVTD→			AmIoU
		FSUW	Caveseg	UVTD	UVTD	FSUW	SUIM	UVTD	Caveseg	SUIM	FSUW	Caveseg	SUIM	
Rein	CVPR'24	36.9	**65.3**	51.8	30.4	11.5	28.1	44.2	**41.7**	48.0	38.9	63.3	63.6	43.6
FADA	NIPS'24	43.7	41.4	56.6	31.6	12.2	28.4	**45.0**	35.6	**51.2**	36.6	48.8	56.2	40.6
SET	ACM MM'24	**53.5**	44.0	54.2	31.8	10.0	**29.8**	43.2	37.4	47.9	**52.1**	59.0	**70.8**	44.5
CDG	PR'25	29.9	34.7	47.6	**32.5**	**16.8**	29.5	43.9	33.3	50.0	32.9	50.9	60.8	38.6
Ours	-	50.0	52.7	**57.5**	31.8	14.8	28.4	43.5	38.2	49.5	48.2	**65.7**	69.4	**45.8**

3) Comparison Using Eva02 Backbone: Table III reports the comparison results of our method against recent state-of-the-art approaches using the EVA02 backbone across various DG scenarios, where each dataset is employed as the source domain in turn. Our method achieves superior performance, obtaining the highest AmIoU of 45.8%, outperforming the second-best method (SET) by 1.3%. Notably, our approach achieves the best or second-best results on nine out of 12 transfer settings, demonstrating strong and stable generalization capability across diverse underwater datasets.

It is worth highlighting that FADA, originally designed to improve generalization in urban scenes under adverse weather conditions (e.g., fog and rain), achieves competitive performance in some settings (e.g., SUIM→UVTD), benefiting from its effective frequency-adapted learning scheme. However, the overall performance of FADA is still substantially lower than our method (AmIoU 40.6% versus 45.8%), indicating that while the challenges in underwater environments share certain characteristics with weather-induced visibility degradation, the complex interplay of scattering, color distortion, and domain shift in underwater imagery requires more specialized modeling strategies.

TABLE IV

COMPARISON WITH SOTA USING MOBILENET-V2 BACKBONE ON SUIM → OTHERS

Method	Venue	SUIM→			AmIoU
		FSUW	Caveseg	UVTD	
IBN	ECCV'18	34.0	20.5	**45.9**	33.5
SW	ICCV'19	32.3	20.7	45.3	32.8
ISW	CVPR'21	33.3	25.6	45.3	34.7
SHADE	IJCV'24	27.1	24.7	36.0	29.3
BlindNet	CVPR'24	32.4	25.4	45.2	34.3
PPAR	TITS'25	31.7	25.8	45.5	34.3
Ours	-	**34.5**	**29.1**	45.4	**36.3**

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TABLE V
COMPARISON WITH SOTA USING MOBILENET-V2 BACKBONE ON UVTD → OTHERS

MR	SUIM→			AmIoU
	FSUW	Caveseg	UVTD	
0	34.0	21.8	45.9	33.9
0.1	33.5	21.9	**46.6**	34.0
0.2	34.0	24.0	46.2	34.7
0.3	34.0	25.9	46.3	35.4
0.4	39.6	**32.7**	41.5	**37.9**
0.5	**39.8**	19.8	42.9	34.2
0.6	39.3	12.6	38.9	30.3

TABLE VI
MR EXPERIMENT ON RESNET-50

Method	Venue	UVTD→			AmIoU
		FSUW	SUIM	Caveseg	
IBN	ECCV'18	40.6	74.5	31.7	48.9
SW	ICCV'19	**41.0**	68.4	33.6	47.7
ISW	CVPR'21	38.2	68.1	31.9	46.1
SHADE	IJCV'24	37.2	71.2	34.3	47.6
BlindNet	CVPR'24	36.9	69.5	28.7	45.0
PPAR	TITS'25	38.5	71.2	33.5	47.7
Ours	-	38.9	**75.5**	**39.6**	**51.3**

4) Comparison Using MobileNet-V2 Backbone: We also investigate the performance of our method under a lightweight backbone, MobileNet-V2. Experiments are conducted using SUIM and UVTD as source domains, and the results are shown in Tables IV and V, respectively. Specifically, when trained on SUIM, our method achieves the highest AmIoU of 36.3%, surpassing the second-best ISW by 1.6% in AmIoU. Meanwhile, our approach achieves the best results on FSUW (34.5%) and Caveseg (29.1%). In UVTD→ others, our method again achieves the highest AmIoU of 51.3%, outperforming both SW and PPAR (47.7%) by 3.6% in AmIoU.

These results demonstrate the effectiveness and generalization ability of our method even under a lightweight backbone. While lightweight backbones typically face challenges in extracting high-level semantic features, our approach remains robust and achieves state-of-the-art performance. This highlights the practical potential of our method for resource-constrained or real-time underwater applications.

C. MR Ablation Experiment

To validate the choice of mask ratio (MR) settings, we conduct ablation experiments under two representative backbone configurations. Specifically, we use ResNet-50 to evaluate the performance under full-parameter training scenarios, applicable to both ResNet-50 and MobileNet-V2 backbones. In parallel, we perform additional experiments with DINOv2 to analyze the behavior under PEFT settings, which are shared by DINOv2 and EVA02.

TABLE VII
MR EXPERIMENT ON DINOV2

MR	SUIM→			AmIoU
	FSUW	Caveseg	UVTD	
0	51.8	50.4	56.4	52.9
0.1	50.4	**52.3**	**58.9**	**53.8**
0.2	46.5	49.6	58.0	51.4
0.3	**51.9**	42.2	57.7	50.6
0.4	48.1	43.7	56.2	49.3

TABLE VIII
ABLATION STUDIES USING RESNET-50 AS BACKBONE UNDER THE SETTING OF SUIM→{FSUW, CAVESEG, UVTD}

DCL	CSMP	FSUW	Caveseg	UVTD	AmIoU	Gain
		28.5	22.8	39.5	30.3	-
✓		34.0	21.8	**45.9**	33.9	+3.6%
✓	✓	**39.6**	**32.7**	41.5	**37.9**	+7.6%

As shown in Table VI, for ResNet-50, increasing MR from 0 to 0.4 progressively improves performance, demonstrating that introducing moderate style-masking effectively simulates realistic degradation and encourages the model to learn robust representations. The best AmIoU is achieved at MR = 0.4 (37.9%), with performance beginning to decline only at MR = 0.5 and higher due to excessive style degradation.

In contrast, as shown in Table VII, for DINOv2, even a small amount of masking already increases modeling difficulty due to the limited parameter update during fine-tuning. Performance peaks early at MR = 0.1, while further increasing MR beyond 0.2 consistently leads to performance drops. Thus, we did not extend the MR beyond 0.4 in this setting.

The selection of MR is determined according to the backbone's adaptation capacity and robustness to style perturbation. For full-parameter training backbones (e.g., ResNet-50), stronger masking can be tolerated due to sufficient parameter adaptability, and thus, a higher initial MR (0.4) is adopted. In contrast, for VFM-based partial fine-tuning backbones (e.g., DINOv2), the limited number of updated parameters makes the model more sensitive to perturbation, so a more conservative initial MR (0.1) is selected.

D. Ablation Studies

To evaluate the effectiveness of each proposed component, we conducted ablation studies. Note that VADM is designed as a feature decoupling mechanism rather than a DG module and, thus, is not included in the ablation comparisons. Moreover, since CSMP is intended to work in conjunction with DCL, experiments focus on two configurations: using DCL alone and using both DCL and CSMP. As shown in Table VIII, the model with DCL alone improves performance by 3.6% over the baseline, and the final model with both DCL and CSMP achieves the 7.6% improvement. These results demonstrate the effectiveness of each component in enhancing generalization performance.

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TABLE IX
ABLATION STUDIES ON AUGMENTATION STRATEGIES

V	A	V&A	FSUW	Caveseg	UVTD	AmIoU
✓			38.6	21.1	43.5	34.4
	✓		39.3	20.6	43.9	34.6
		✓	35.8	25.0	46.4	35.7
✓	✓		33.5	27.0	46.9	35.8
✓		✓	33.7	27.7	46.6	36.0
	✓	✓	38.7	25.5	45.5	36.6
✓	✓	✓	39.6	32.7	41.5	**37.9**

TABLE X
ABLATION STUDIES ON LOSS WEIGHT

λ_1	λ_2	FSUW	SUIM	Caveseg	AmIoU
0.001	0.001	53.1	71.2	71.3	65.2
0.0001	0.001	64.3	60.2	75.3	66.6
0.00001	0.001	54.7	71.4	72.6	66.2
0.0001	0.01	54.2	72.5	71.3	66.0
0.0001	0.0001	62.5	60.2	72.3	65.3

Moreover, we conduct ablation experiments on visibility augmentation (V), appearance augmentation (A), and their combination (V&A). As shown in Table IX, when all three strategies are combined, the performance reaches the best result. These results validate that both visibility and appearance augmentations are complementary and jointly beneficial for enhancing model generalization in an underwater environment.

E. Ablation Studies on Loss Weight

We conducted additional ablations on the loss weights λ_1 and λ_2, as shown in Table X. The results indicate that moderate values ($\lambda_1 = 0.0001$ and $\lambda_2 = 0.001$) achieve the best tradeoff across datasets, yielding the highest AmIoU of 66.6%. Small or large weights can degrade performance on certain domains, confirming that both losses contribute positively but should be balanced for optimal cross-domain segmentation.

F. Feature Visualization

As discussed above, A and t are interpreted as the appearance and visibility. To better understand the roles of the predicted A and t, we visualize their feature maps in Fig. 7. As described in Implementation Detail of Predictors P_A and P_t, these two terms are derived from a learned transformation path between the original feature f and the stylized feature f' via an attention-guided differential fusion, where the feature pair (f and f') shares the same content but differs in the whole style. We observe that A exhibits spatially continuous and high-response regions that generally align with major semantic structures, such as large foreground objects or uniform backgrounds. Compared with the low-response areas, these highlighted regions tend to possess richer textures and more complex appearance characteristics. Notably, only a portion of the background is activated, which is reasonable. On the one hand, some background areas contain textured structures

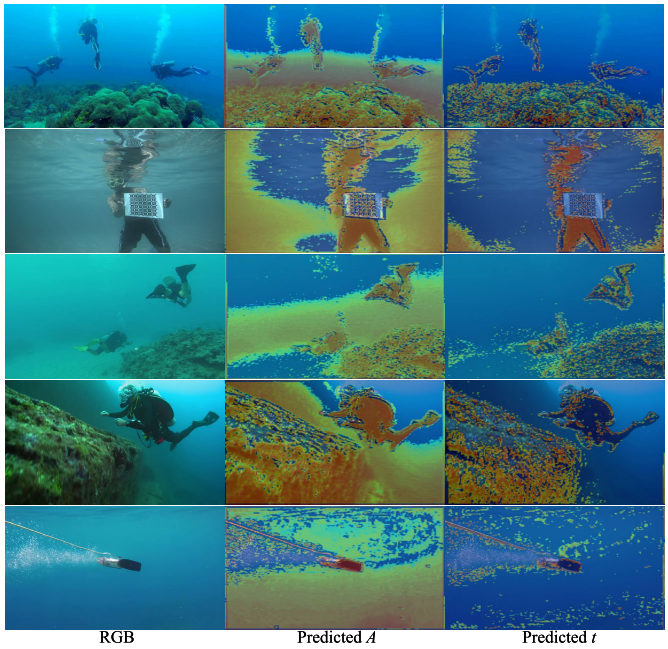


Fig. 7. Feature visualization of predicted A and predicted t.

that significantly contribute to the image's overall style and are, thus, activated. On the other hand, flat and textureless background regions contribute little to appearance modeling and are naturally suppressed. In summary, the high-response regions of A align with texture-rich areas that define the overall appearance, supporting the interpretation of A as an appearance-related component. In contrast, the predicted t displays highly localized and scattered responses with fine-grained spatial variations. First, t exhibits stronger responses in near-field regions, which is consistent with the ASM: t represents the transmission ratio of light from the object to the camera. A lower t indicates stronger scattering and blur, while a higher t suggests clearer visibility. As foreground regions are generally closer and more visible, their higher response is expected. Second, we observe strong responses around object boundaries. This is reasonable because visibility often changes most drastically near edges. These regions typically exhibit sharp transitions in clarity, leading to higher variations in t and thus stronger activations. This further confirms that t is effectively capturing visibility information.

Overall, the visual patterns of A and t strongly support the effectiveness and interpretability of the proposed VADM method.

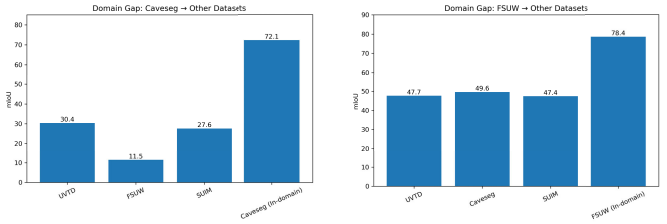
G. Trainable Parameters in VFM-Based Backbone

We analyzed the trainable parameters under the VFM-based backbone. As shown in the Table XI, our proposed framework introduces only 4M additional trainable parameters compared to the baseline while achieving a 1.1% AmIoU improvement on the UVTD→other datasets. Moreover, according to Table II, our method outperforms the baseline by 2.1% on average, demonstrating that the performance gain comes at a minimal training cost. Regarding runtime, all additional modules (ASM, AdaIN, and style masking) are used only

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TABLE XI
COMPARISON ON TRAINABLE PARAMETERS T_p

Methods	T_p	UVTD→			AmIoU
		FSUW	Caveseg	SUIM	
Rein [40]	2.99M	58.9	63.0	74.5	65.5
FADA [41]	11.65M	53.0	66.1	62.7	60.6
SET [44]	6.13M	56.8	57.7	73.1	62.5
CDG [42]	2.99M	41.8	40.2	58.2	46.7
Causal-Tune [67]	7.55M	57.3	51.5	71.9	60.2
Ours	6.99M	64.3	60.2	75.3	66.6



(a) (b)

Fig. 8. Quantitative analysis of the domain gap under single-source training using DINOv2. (a) Caveseg → Others. (b) FSUW → Others.

during training. During inference, these modules are not involved, so the testing time remains identical to the baseline, ensuring that the practical deployment cost is unaffected. This design allows the framework to improve generalization while maintaining efficient inference.

H. Domain Gap Analysis

To explicitly quantify the magnitude of the domain gap, we conducted additional cross-domain baseline experiments using DINOv2 under single-source training settings. As shown in Fig. 8, when training on Caveseg, the in-domain performance reaches 72.1% mIoU, while the cross-domain results drop dramatically to 30.4% (UVTD), 27.6% (SUIM), and especially 11.5% (FSUW). This corresponds to a degradation of up to 60.6% mIoU, clearly indicating a severe domain discrepancy. Similarly, when training on FSUW, the in-domain performance is 78.4% mIoU, whereas cross-domain performance decreases to 47.7% (UVTD), 49.6% (Caveseg), and 47.4% (SUIM), showing a consistent gap of approximately 30% mIoU. These significant performance drops demonstrate that the domain gap among underwater datasets is substantial and nontrivial. The variation in degradation magnitude across dataset pairs further suggests heterogeneous domain discrepancies.

I. Failure Case Analysis

Although the proposed framework shows strong generalization ability, it still exhibits notable failure modes when objects appear only partially at the image boundary, as shown in Fig. 9. First, the proposed model lacks explicit shape- or boundary-aware perceptual learning (like [29]), making it difficult to infer object identity from small, truncated fragments with incomplete contours. Second, the visibility–appearance

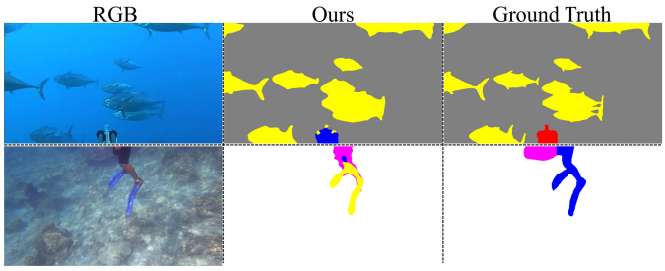


Fig. 9. Some failure cases.

decoupling is performed at the global image level rather than in a class-specific manner; thus, when only a small portion of an object is exposed, the absence of category-discriminative cues further amplifies interclass confusion. These two factors jointly lead the network to misinterpret the visible fragment as background or confuse it with semantically similar categories. Future work may incorporate boundary-aware priors, class-specific modeling, or structure-completion strategies to mitigate such issues.

VI. CONCLUSION

In this article, we introduced UnderwaterDG, a new task that formally defines DGSS in underwater environments. Motivated by the observation that both visibility and appearance variations contribute significantly to domain shifts in underwater scenes, we proposed a novel framework for learning visibility- and appearance-invariant representations. Specifically, we proposed a novel VADM method to disentangle appearance and visibility. Furthermore, we introduced a DCL to promote invariance across visibility and appearance dimensions and a CSMP strategy to enhance model robustness through progressive perturbations. Comprehensive experiments across multiple underwater datasets demonstrate that our method significantly outperforms existing approaches and shows strong generalization to unseen marine scenes.

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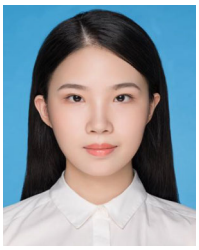
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